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Gold nanodoughnut as an outstanding nanoheater for photothermal applications

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Supplementary material

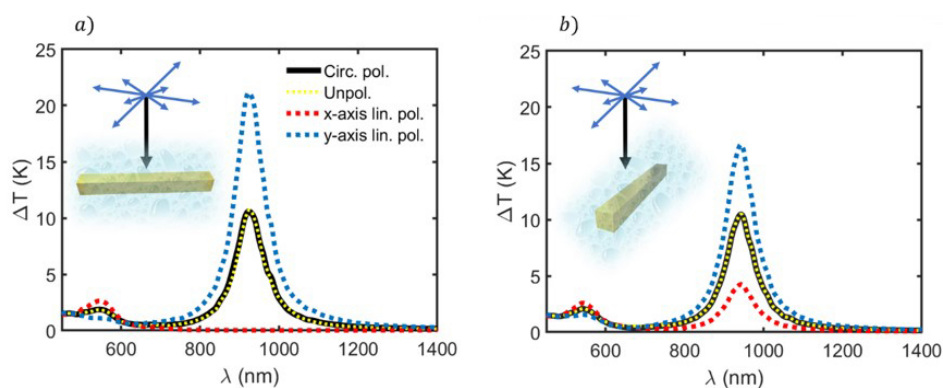


Fig. S1: Thermal response in water of a perfectly oriented rectangular rod (a) and a 30° x-axis and 60° y-axis rotated rod (b) excited by x-polarized (red dotted line), y-polarized (blue dotted line), circular (black solid line) and unpolarized light (yellow dotted line). The dimensions of the rectangular rod are 120 nm length with a section length of 30 nm and for the toroid has a main radius of 50 nm and a second radius of 10 nm.

Fig. S1 justifies the use of circularly polarized light in this work. It is used to emulate the influence that the unpolarized illumination would have on the particle thermal responses. The most common way of simulating unpolarized light consist in performing two different simulations for orthogonal input electric fields and taking the mean of the results of such simulations. However, executing only one simulation considering circular polarization, results in a higher computational efficiency. As can be seen, the rectangular rod thermal responses are equivalent for circular and unpolarized light, both deriving in a dramatic thermal efficiency reduction with respect to the most favourable y-axis linear polarized light.

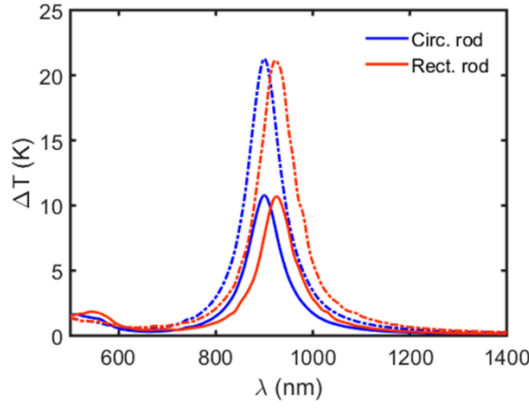


Fig. S2: Spectral response comparison of circular (blue) and rectangular (red) rod in water for linear (dotted line) and circular (continuous line) polarizations. The dimensions of the rectangular rod are 120 nm length with a section length of 30 nm. The circular rod has the same length and the needed radius to remain the volume constant.

Fig. S2 has been constructed to justify the election of the rectangular rod. The circular and rectangular rod thermal responses show similar maximum temperature increases for both incident polarizations, reducing its thermal efficiency for circular polarization as shown above. It can be seen that rectangular rod response is slightly redshifted with respect to the circular one making them equivalent. As the circular geometry corresponds to the maximal cross section for a fixed volume, the aspect ratio (minor length over major length) is increased with respect to the squared rod, explaining the observed blueshift.

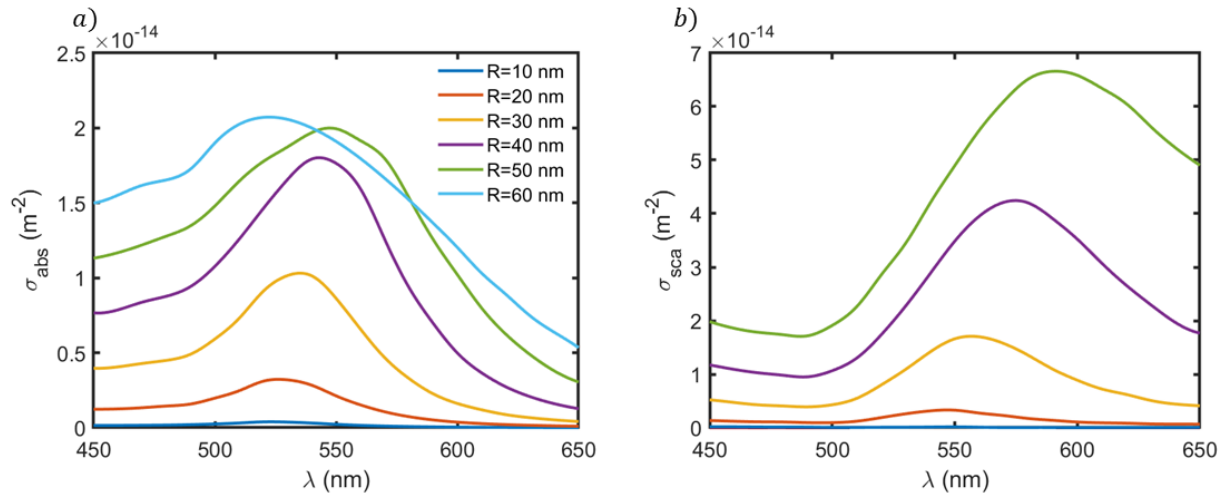


Fig. S3. Absorption (a) and scattering (b) cross-sections of an isolated gold sphere in water for a set of radii.

In Fig. S3, the scattering and absorption cross-sections for a gold sphere in water are shown. It can be seen, the well-known redshift in the scattering spectra as a result of the diameter increase. The absorption cross section also presents a slight redshift for the smallest particles.

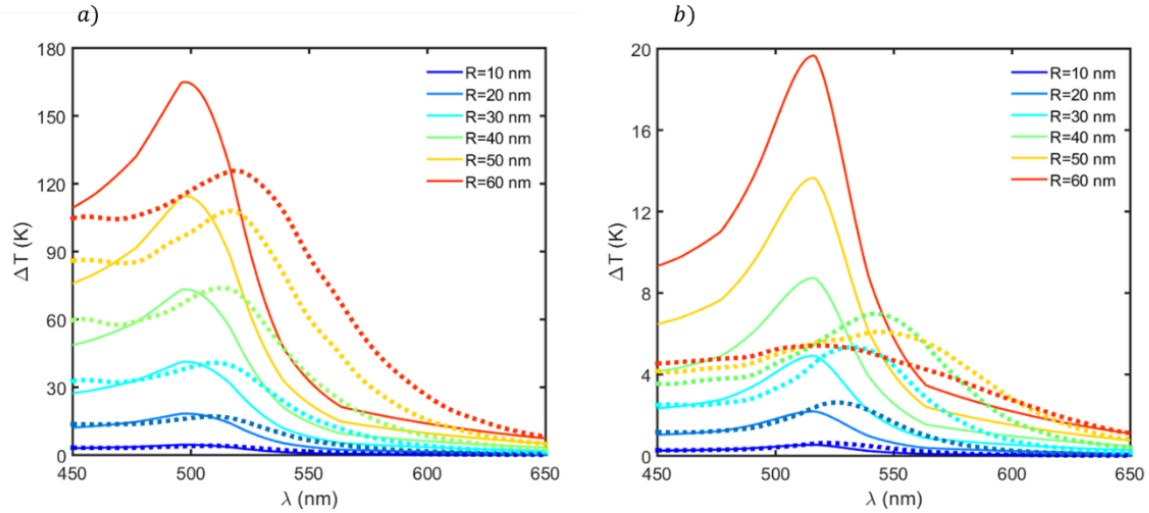


Fig. S4: Simulated spectral thermal response of a sphere (dotted line) compared with the obtained according to the expressions used by Govorov and co-workers (solid line) in air (a) and in water (b). Radii between 10 and 60 nm have been studied.

Fig. S4 shows the spectral thermal response of a sphere in air and in water for radii between 10 and 60 nm. In the last case, the obtained temperatures are much lower than in air due to the higher thermal conductivity of water. Attending to the Govorov calculations, the increase of temperature grows with the radius, keeping the resonance wavelength almost constant for both surrounding fluids and being redshifted in water with respect to air as can be expected from Mie theory. Although our simulations in air show a good agreement between curves for lower radii, it can be seen how a huge discrepancy appears for higher values, being the simulated maximum of temperature lower than the corresponding one given by the Govorov's expressions. This significant difference can be explained in terms of the relative size of the excitation wavelength and the particle. For higher radii, the quasi-static approximation gets weaker, avoiding the description of the sphere as a perfect electric dipole. Furthermore, a possible thermal explanation could come from the resistive losses which are concentrated in a smallest and superficial region reducing the temperature increase at the particle in such a way that the relation between maximum temperature increment and particle size becomes more complex. Spheres show a poor spectral tunability for both surrounding mediums. In the following, we will show a comparison of all the analysed geometries (gold/silica core-shells, stars, disks, rods and toroids) in air together with some of the most representative nanoparticle designs for circular polarization, demonstrating the impact that every topological parameter has on its thermal behaviour.

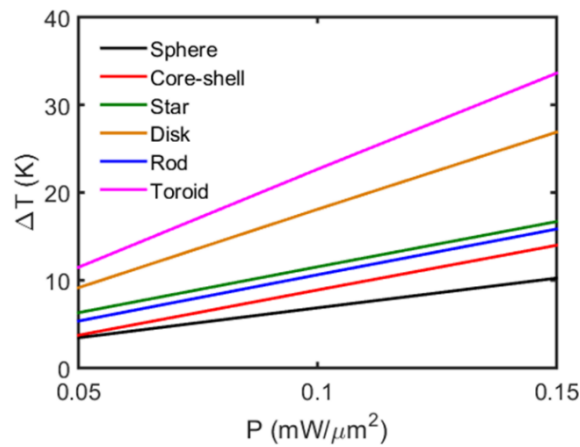


Fig. S5: Maximum temperature increase of the selected structures in Fig. 2 illuminated with circular polarized light as a function of the incident power density.

Fig. S5 shows how the power density affects the thermal efficiency of the nanostructures, showing that the temperature can be enhanced by increasing the power density. Noticeable differences between curves appears. Focusing on the slopes, it can be observed that disks and toroids present the fastest growth of temperature for higher power densities being the toroid one superior in contrast with the rest of structures. This can be explained in terms of the particles volume. Since spheres, core-shells, stars and rods are bigger than toroids and disks for the optimal dimensions shown in Fig. 2, they present a weaker growth with respect to the power density.

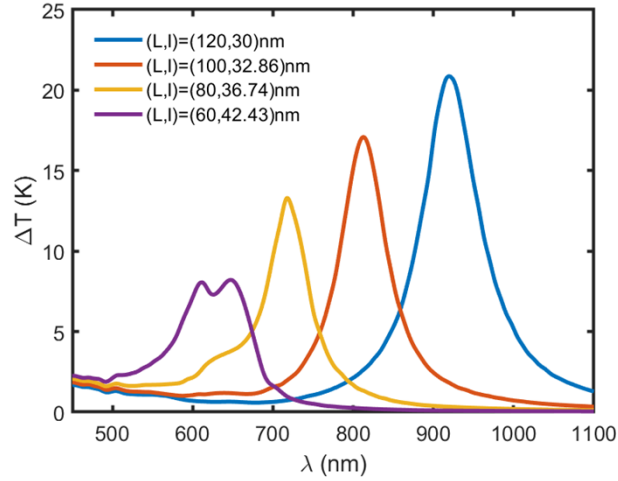


Fig. S6. Spectral thermal response for a set of gold rods with identical volumes. The largest dimension is limited to 120 nm.

Fig. S6 shows the spectral thermal response of gold rods with different sizes remaining their volume constant. It can be seen how the maximum temperature increment is decreased as the largest rod dimension reduces. This can be explained in terms of volume: since it must be constant, the shortest dimension of the rod increases for a decreasing larger dimension reducing the resistive losses and the rod heating ability consequently.

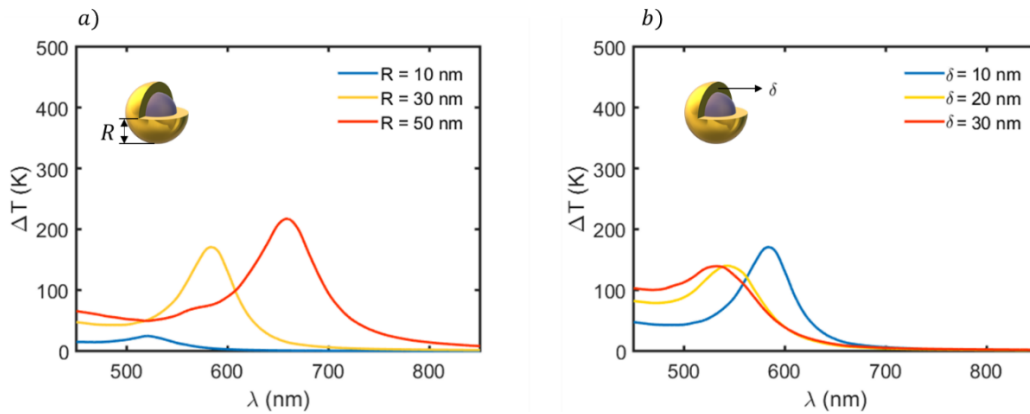


Fig. S7: Gold/silica core-shell simulation for variations in the core radius (R) and the shell thickness (δ) in air. a) Core-shell thermal response for core radius between 10 and 50 nm for a given shell thickness of 10 nm. b) Core-shell thermal response for shell thickness between 10 and 30 nm for a fixed core radius of 30 nm.

Fig. S7 shows the spectral thermal response of a core-shell structure and how the silica core makes it redshift with respect to the single material sphere (Fig. S4), pushing the system closer to the biological window. In Fig. S7a, the effect of the core radius is clearly demonstrated: as the core size increases for a fixed shell thickness, the spectral response is redshifted and the maximum temperature grows since the shell expansion

carries to an increment of the resistive losses. On the other hand, if the core radius is fixed, the structure seems more like a sphere for thicker shells, giving rise to a reduction in the maximum temperature and a blueshift of the spectral response.

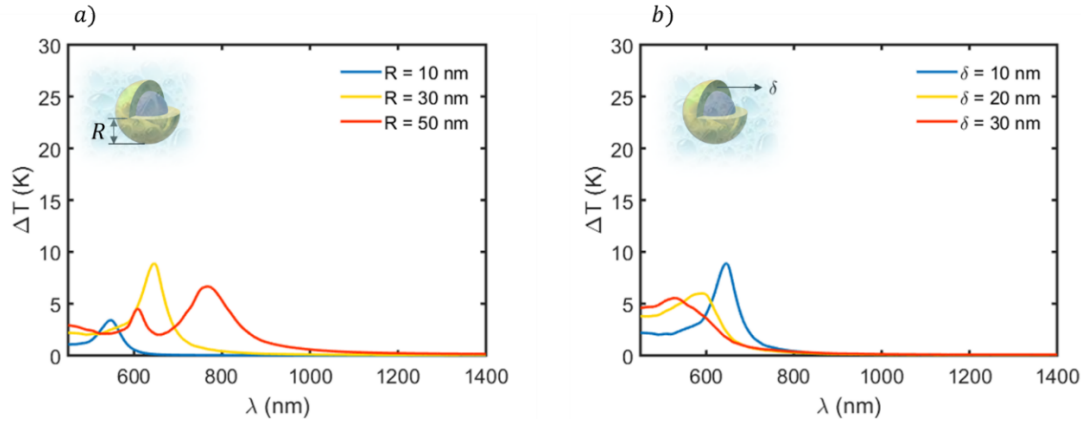


Fig. S8: Gold/silica core-shell simulation for variations in the core radius (R) and the shell thickness (δ) in water. a) Core-shell thermal response for core radius between 10 and 50 nm for a given shell thickness of 10 nm. b) Core-shell thermal response for shell thickness between 10 and 30 nm for a fixed core radius of 30 nm.

Fig. S8 shows the spectral response of a core-shell in water for different core radii and shell thicknesses. As in the case of air, in Fig. S8a, it is shown how as the core size increases for a given shell thickness, the spectral response is redshifted for increasing core radius. Conversely, as plotted in Fig. S8b, for a fixed radius, the structure seems more like a sphere for thicker shells, leading to a reduction in the maximum temperature and a blueshift of the spectral response.

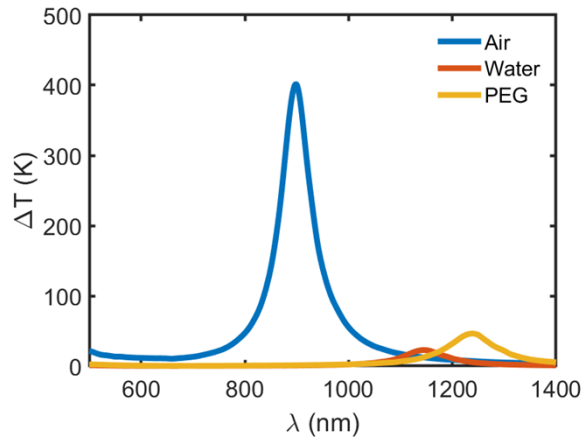


Fig. S9: Comparison of the spectral thermal response of a toroid immersed in air (blue), water (red) and PEG (yellow) for circular polarization. The dimensions of the toroid are $(R,r) = (50, 10)$ nm.

Fig. S9 shows the thermal spectral response of a toroid immersed in three different fluids: air, water and PEG. It can be seen the huge discrepancies between the optimum temperature increase for the three media, reaching temperatures of 400 K in air, 23 K in water and 45 K in PEG approximately. This can be explained in terms of their thermal conductivity: 0.02 W/mK for air, 0.6 W/mK in water and 0.3 W/mK for PEG. Thus, larger temperature increments are obtained for less thermal conductive media. Another remarkable variable to consider is the refractive index, the PEG being the most refringent. These contrasts lead to the redshifts in Fig. S9.

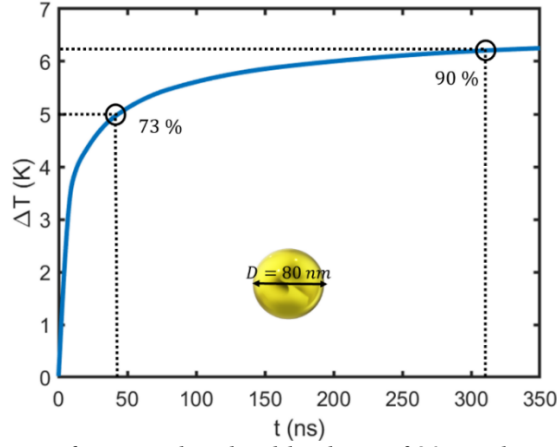


Fig. S10: Transient thermal state for an isolated gold sphere of 80 nm diameter in water illuminated with unpolarized light at 540 nm.

Fig. S10 shows the thermal transient state of a gold sphere in water illuminated at 540 nm. As can be seen, if we consider the criterion followed in this work, the sphere takes 310 ns to reach the 90% of the stationary temperature. However, this value varies depending on the selected criteria. For instance, if we consider a logistic growing, then the characteristic time will be reached at 73% of the stationary temperature approximately. Thus, the characteristic time would be 45 ns as seen in the figure, which fits with the estimated values from $t \approx D^2/\alpha$ considering the diameter as characteristic length. Thus, the temporal values depend on the selected criterion without affecting the calculations suitability. In our work, the 90% of the stationary temperature was selected as the conductive criterion.

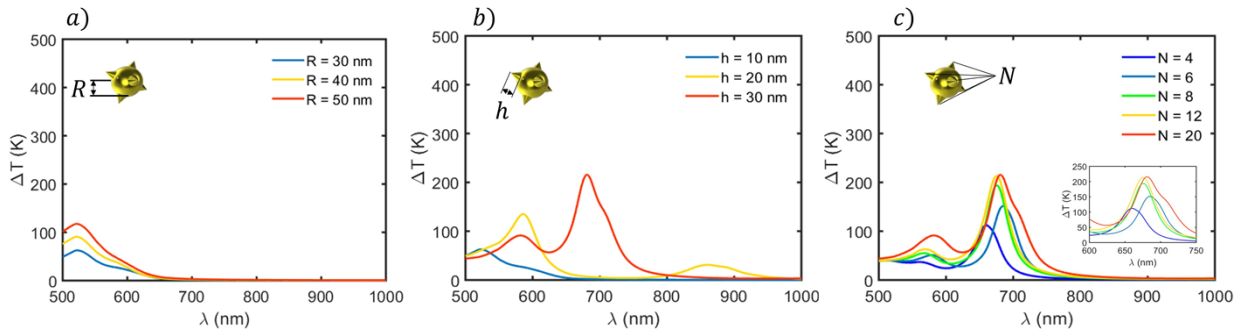


Fig. S11: Star simulations for a varying core radius (R), cone height (h) and number of cones (N) in air. a) Thermal spectrum for a star geometry with 20 cones of height 10 nm. b) Thermal spectrum for a star with 20 cones and a core radius of 30 nm. c) Thermal spectrum of a star geometry with a core radius of 30 nm and a variable number of cones of 30 nm height.

Fig. S11 demonstrates the tunability of nanostars, an alternative approach to improve the thermal response of nanoparticles. Attending to Fig. S11a, we can see how the thermal response is weakly redshifted with respect to the core-shell structure and a secondary resonance appears below 600 nm. The dual resonance observed for the nanostar is due to this intricate geometry: the smaller peak corresponds to a resonance of the spherical core, while the most prominent corresponds to a whole-structure resonance. As shown in Fig. S11b, if the cones height is reduced, the second resonance weakens and the nanostar approaches the single sphere thermal response.

The highest temperature increment obtained with this structure is 200 K. Although the nanostar structure gives a better thermal response than in the case of core-shell and sphered particles for an equivalent temperature increment and wavelength region, the structure is highly regular and the response is extremally sensitive to the number of cones and its sizes as can be seen in Fig. S11c. Hence, the nanostar fabrication

requires high precision, becoming quite challenging. Due to this, the resulting structures are often irregular in such a way that calculations considering regular geometries are speculative.

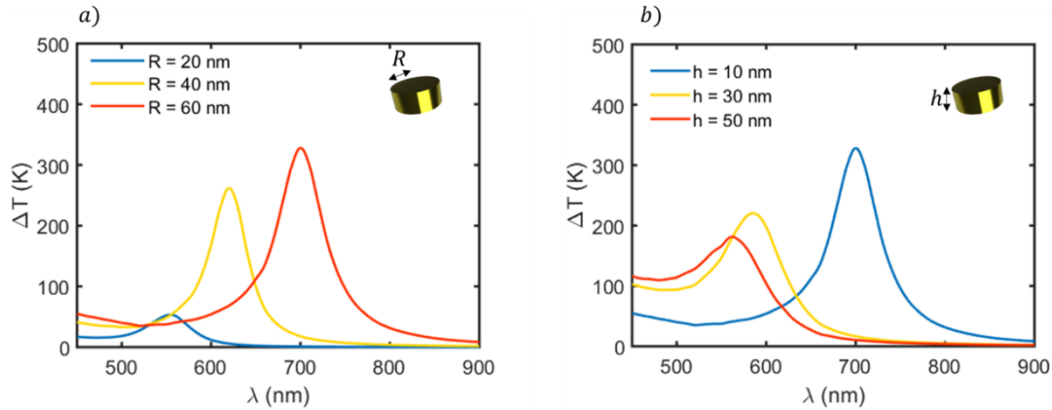


Fig. S12: Disk simulations for a varying radius (R), and height (h) in air. a) Thermal spectrum for a disk geometry of height 10 nm. b) Thermal spectrum for a disk with 60 nm radius.

Fig. S12 shows the spectral thermal response of the nanodisk structure for a set of representative dimensions. It can be seen in Fig. S12a how, as the radius increases, the thermal response is redshifted and higher temperature increments are obtained. The reason for this is that a radius enlargement leads to a growth in the electronic motion amplitude, giving rise to such a redshift and temperature increment. In addition, as Fig. S12b plots, a reduction of the disk height produces an augmentation of the temperature increment, as the structure resistance is increased. In case of the optimal disk (10 nm height, 60 nm radius), the maximum temperature enhancement is 320 K approximately.

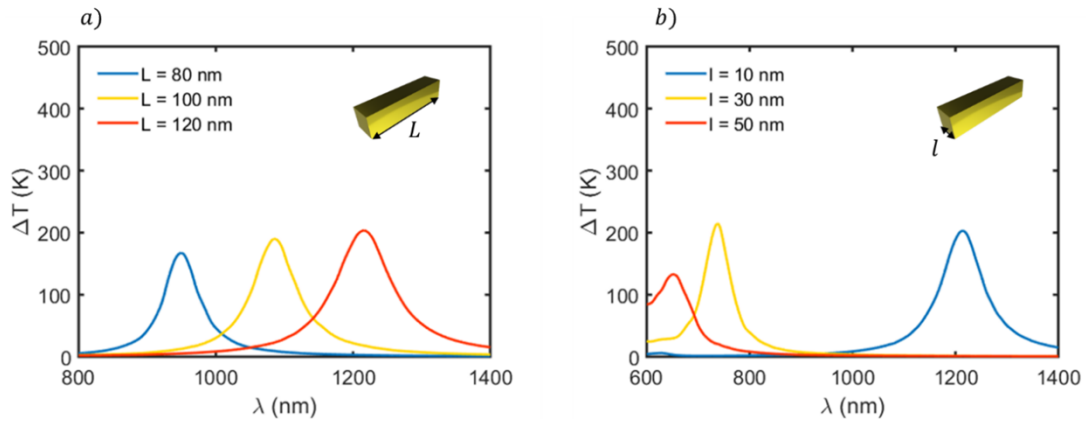


Fig. S13: Rod simulations for a varying major length (L) and section length (l) in air. a) Thermal spectrum for a rod geometry of $l=10$ nm. b) Thermal spectrum for a rod of $L=120$ nm.

Fig. S13 shows the spectral thermal tunability of the rectangular rod geometry. In Fig. S13a, it can be seen how as the major dimension increases, electrons within the material will oscillate with higher amplitudes, resulting in a larger excitation wavelength. In turn, as the section size increases and approaches the major dimension, the geometry will be akin to a sphere, resulting in a resonance blueshift (Fig. S13b). The rod response to circularly polarized light can be explained by considering the rotating nature of the circular polarization: as the electric field rotates, the major dimension of the rod and the electric field will only concur at specific times, at which a dipolar mode for the adequate wavelength is generated. Conversely, at successive stages, they will be crossed in such a way that the previous mode will disappear, resulting in a lower maximum temperature increment of around 200 K.

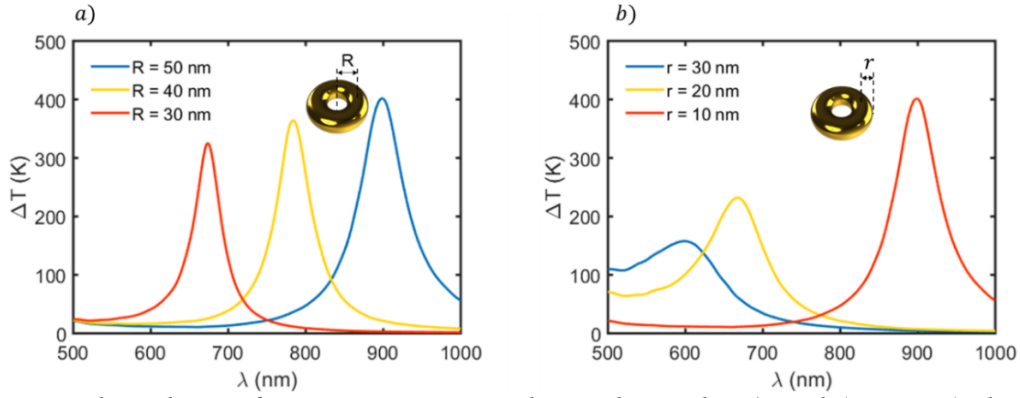


Fig. S14: Toroid simulations for a varying main and secondary radius (R and r) in air. a) Thermal spectrum for a doughnut of 10 nm secondary radius. b) Thermal spectrum for a doughnut of 50 nm main radius.

Fig. S14 shows the spectral tunability of the toroid which is the last studied shape to improve the nanoparticle thermal response. A maximum temperature increment of around 400 K is obtained for the optimal toroidal geometry (50 nm major radius, 10 nm minor radius). In Fig. S14a it can be seen, how as the major radius increases, temperature increments occur for higher excitation wavelengths, as a result of the interaction between the two toroid sides. If the main radius grows, the induced dipoles move away leading to a strong redshift which gives the toroid a huge tuning capability. On the other hand, as plotted in Fig. S14b, if the secondary radius increases, the opposite effect can be observed: as this radius increases, the induced dipoles get closer, resulting in a significant blueshift. This clearly demonstrates the superior tunability of toroidal shaped nanoparticles over the rest of structures. The previous study has been performed considering the nanostructures immersed in water to expose the effect that the topological parameters have on the spectral thermal response of each geometry.

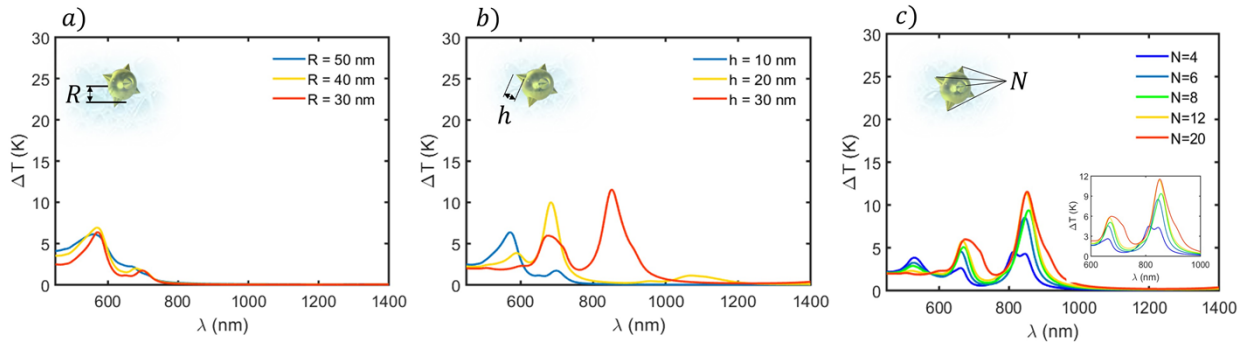


Fig. S15: Star simulations for a varying core radius (R), cone height (h) and number of cones (N) in water. a) Thermal spectrum for a star geometry with 20 cones of height 10 nm. b) Thermal spectrum for a star with 20 cones and a core radius of 30 nm. c) Thermal spectrum of a star geometry with a core radius of 30 nm and a variable number of cones of 30 nm height.

Fig. S15 shows the tunability of nanostars in water. Attending to Fig. S15a, we can see how the thermal response, generally, is weakly redshifted with respect to the core-shell structure. The evolution of the thermal response upon modification of the topological parameters exhibits a similar behaviour to the air medium case.

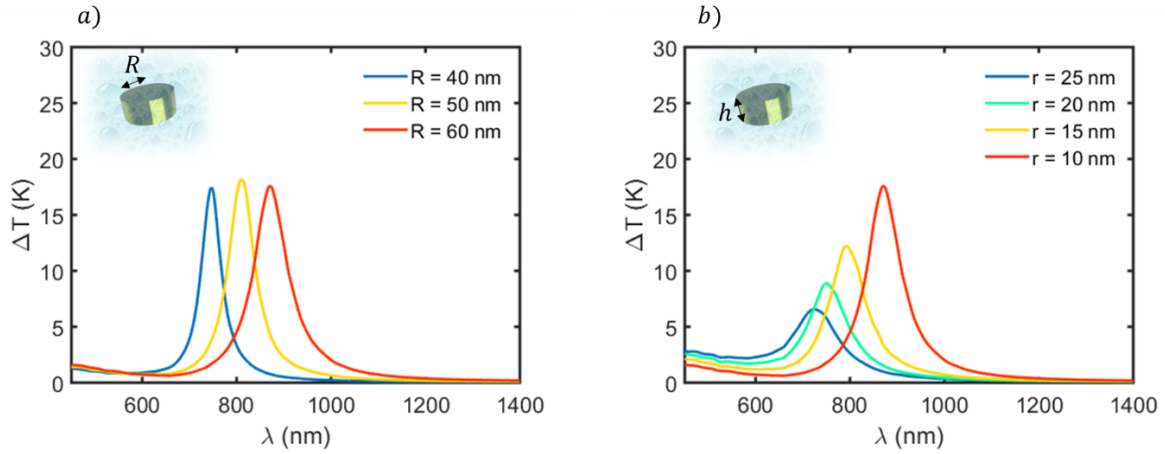


Fig. S16: Disk simulations for a varying radius (R), and height (h) in water. a) Thermal spectrum for a disk geometry of height 10 nm. b) Thermal spectrum for a disk with 60 nm radius.

Fig. S16 shows the tunability of disks in water. Attending to Fig. S16a, we can see how the thermal response shows similar temperature increments while the usual redshift is observed for larger radii. Similarly to the air case, reduction of the disk height result in a better thermal performance as can be seen in Fig. S16b.

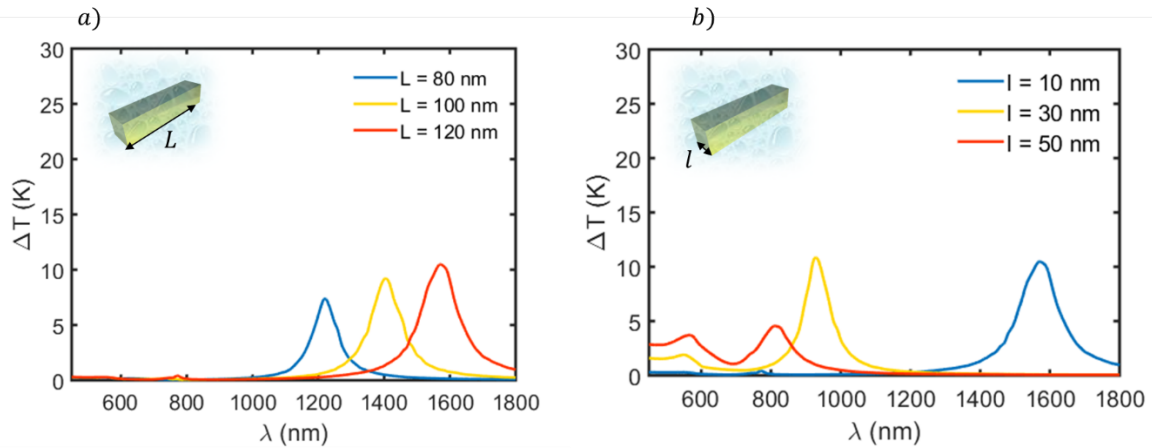


Fig. S17: Rod simulations for a varying major length (L) and section length (l) in water. a) Thermal spectrum for a rod geometry of $l=10$ nm. b) Thermal spectrum for a rod of $L=120$ nm.

Fig. S17 shows the spectral thermal tunability of the rectangular rod geometry in water. In Fig. S17a, although redshifted, the thermal spectrum shows a similar behaviour to the air calculations when the structure dimensions are modified. However, as depicted in Fig. S17b, a new optimal configuration, with a 30 nm section length, appears.

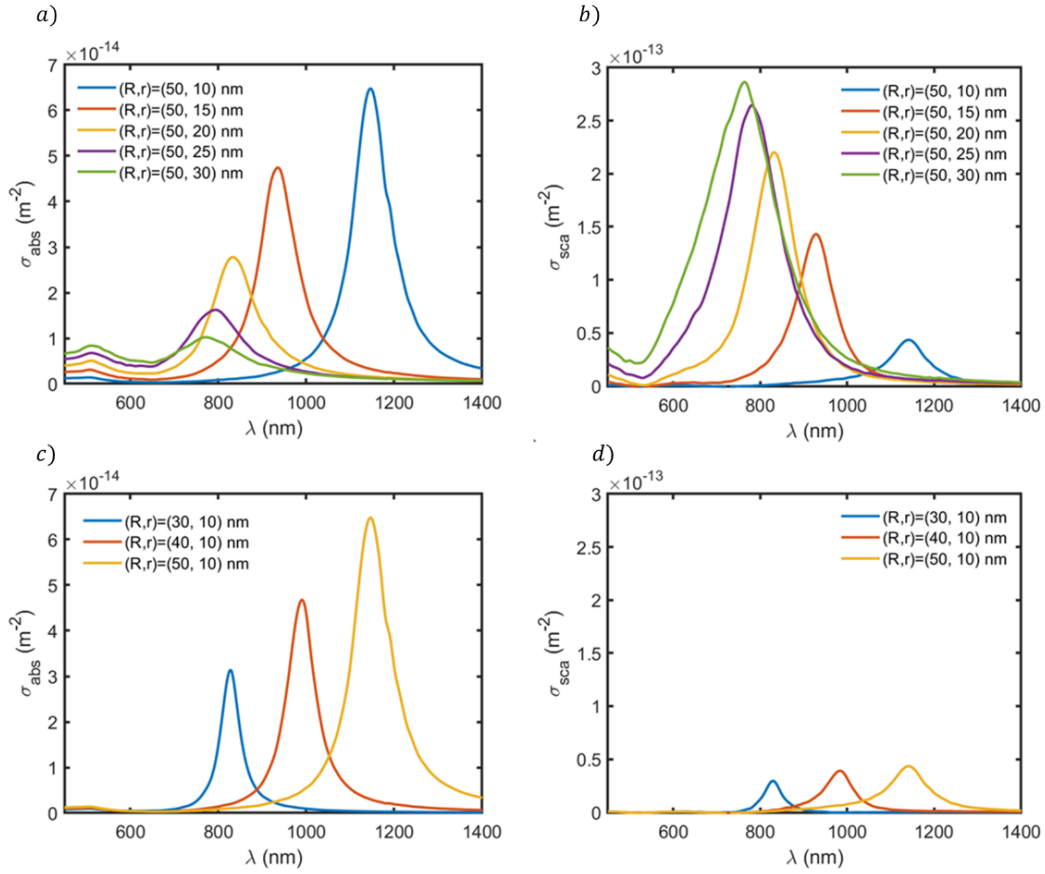


Fig. S18. Absorption (a, c) and scattering (b, d) cross-sections of gold nanodoughnuts in water for different sizes, ranging from 30 to 50 for the main radii and 10 to 30 nm for the secondary one.

Fig. S18 shows the absorption and scattering cross-sections for a set of gold nanodoughnuts. In (a) and (b), the main radius remains constant for a varying secondary radius revealing an interesting behavior: for increasing secondary radius, the spectral response is redshifted and becoming the toroid a better absorber, which fits with the temperature results shown in Fig. S19. In contrast, the scattering cross-section is reduced for growing secondary radius. On the other hand, in (c) and (d), the same pattern can be seen in the absorption cross-section: the absorption increases as the main radius grows. Besides, the scattering cross-section remains almost constant, disregarding the well-known redshift.

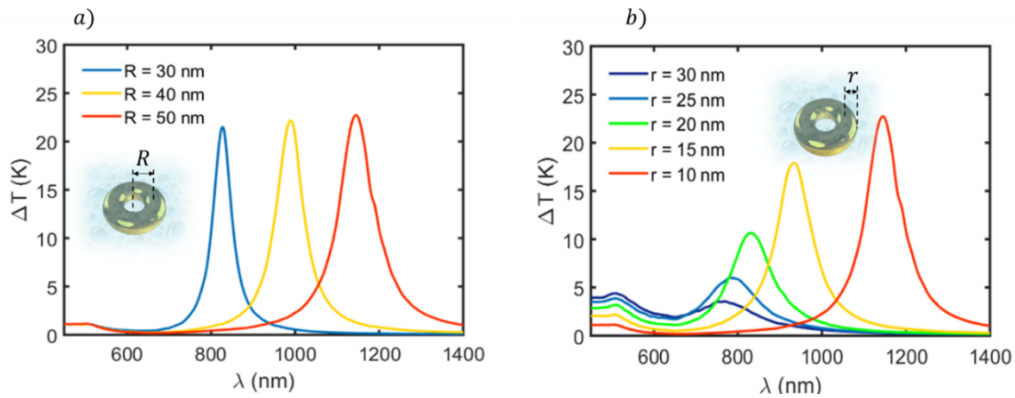


Fig. S19: Toroid simulations for a varying main and secondary radius (R and r) in water. a) Thermal spectrum for a doughnut of 10 nm secondary radius. b) Thermal spectrum for a doughnut of 50 nm main radius.

Fig. S19 shows the tunability of toroids in water exhibiting a behaviour fully parallel to its air counterpart. Focusing on the target wavelength region (second biological window, 1000-1400 nm), it can be seen how the 50 nm major radius and 10 nm minor radius configuration is the only one capable of reaching that spectrum.

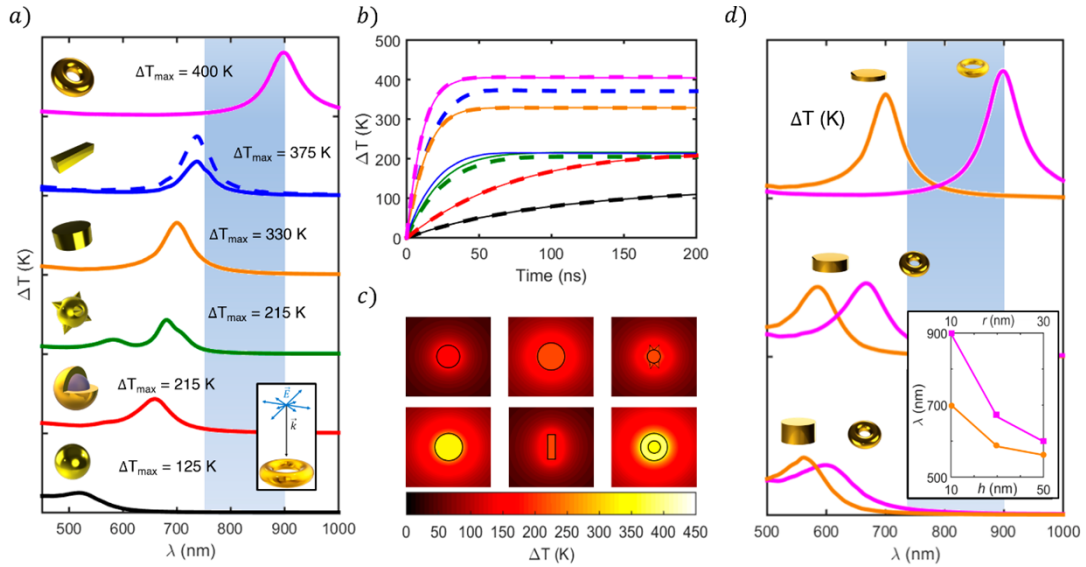


Fig. S20: Gold nanoparticle shape comparison in air for linear (dotted line) and circular (continuous line) polarizations at normal incidence. The thermal responses of spheres (in black), gold/silica core-shells (in red), stars (in green), disks (in orange), rods (in blue) and toroids (in pink) are represented. a) Optimal thermal response for each analysed geometry considering a major dimension lower than 120 nm are plotted for temperatures between 0 and 500 K. The sphere has a radius of 60 nm, cores-shells have a core radius of 50 nm and a shell thickness of 10 nm. The core of stars has a radius of 30 nm with a cone height of 30 nm. Disks radius is 60 nm, with a height of 10 nm. The dimensions of the rectangular rod are 30x120x30 nm. Finally, the toroid has a main radius of 50 nm and a secondary radius of 10 nm. The blue shadowed region represents the first biological windows. b) Transient optimal-point heating for the shapes shown in a) in the corresponding colours. c) Temperature increase spatial distribution for the studied shapes in a). d) Tunability comparison of disks and toroids represented between 0 and 500 K. The disk radius is fixed at 60 nm and the torus main radius at 50 nm. The disk heights are 10, 30 and 50 nm and the torus secondary radii, 10, 20 and 30 nm. Relative wavelength displacement is plotted in the inset.

Fig. S20 shows the photothermal spectral response of the set of the different optimized structures considering its steady (a) and time dependent thermal response (b). Particles are in air being irradiated at normal incidence for circular and linear polarizations. As in Fig. 2, the three upper shapes (disk, rod and toroid) clearly outclass the lower three (sphere, core-shell and star) in terms of temperature generation. Spherical geometries are the most extended particles due to its simplicity, however, they show the worse thermal response offering temperature increments of $\Delta T \approx 125$ K as can be seen in Fig. S20a. These increments reach their maximum in a wavelength far from the biological windows. Meanwhile, the gold/silica core-shell although located in a more usable wavelengths; still shows a poor temperature increase $\Delta T \approx 215$ K compared with the upper three shapes. Finally, the nanostar geometry shows a similar behaviour to the core-shell, with an optimal temperature increment of around 215 K at the same excitation wavelengths.

On the other hand, as in the case of water, the disk, rod and toroidal shapes demonstrate higher temperature increases with respect to the others. The rod response exhibits a redshift compared with the disk. As can be seen, a huge discrepancy between linear and circular polarization responses appears. The linear polarization illumination (input electric field polarized along the greatest dimension of the rod), gives a temperature increment peak of 375 K, higher than the disk one. However, for circular polarization, a considerable reduction in the temperature increment can be observed, resulting in a maximum temperature increment of around 200 K, closer to the core-shell response. Finally, in the case of the toroidal geometry, a maximum temperature increment of around 400 K can be seen for the optimal geometry, equivalent for both linear and circular polarizations.

Figure S20b shows the results of the thermal time-dependent calculations at the optimal wavelengths depicted in Fig. S20a. The three worst particles (sphere, core-shell and star) require also longer times to reach the steady state, up to around 200 ns for the sphere and core-shell, the star having a significantly shorter time of 100 ns. In contrast, the three best heating candidates' shapes display steady state-reaching times as short as 50 ns, furthering the case for these three nanoparticles. Furthermore, the toroidal particle displays a rapid heating, reaching its maximum temperature after a short 30 ns illumination, being the fastest heater.

The temperature spatial distribution shown in Fig. S20c briefly summarizes the thermal behaviour of the studied shapes. As it is shown, disk and toroidal particles offer very high temperatures for circular polarization and heats up the greatest surrounding volume, compared with the rest of structures. Toroids also show great temperatures in their inner holes, outside the material while all geometries display their maximum temperature inside gold.

Another essential aspect to be considered is the particles spectral tunability. This has been compared in Fig. S20d for disks and toroids, which are the two best candidates due to their excellent thermal response and polarization non-dependence. Starting from two designs with similar thermal responses, the characteristic parameters (height for the disk and inner radius for toroids) have been varied. It can be seen how for equivalent geometrical variations, we can get larger wavelength shifts demonstrating the exceptional tunability of toroids.

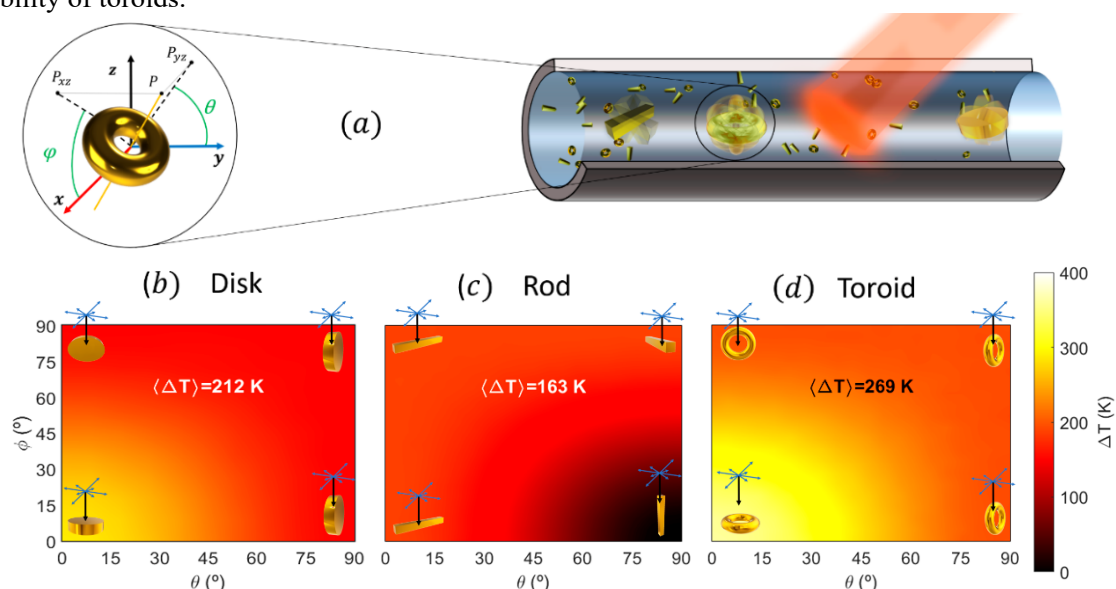


Fig. S21: Thermal responses of the three best nanoheaters in Fig. S20 to rotations. a) Illumination and rotation outline. θ and ϕ are the rotations with respect to the x and y axes and the considered power density is $0.1 \text{ mW}/\mu\text{m}^2$. Maximum temperature increase for linearly polarized and unpolarized light as a function of rotation angles: b) Disk; c) Rod; d) Toriod. The average temperature is shown for all cases. Top insets: linear polarization, bottom insets: unpolarized light.

In Fig. S21, the implications of the particle orientation in the thermal response of the aforementioned structures are clearly demonstrated in air. The disk colormap in Fig. S21b, shows a certain revolution symmetry, as the revolution axis of the disk and the z axis (beam direction) match in the initial configuration. Thus, considering the temporal evolution of the electric field vector for this polarization, rotating the disk about the y axis, and subsequently about the x axis, has the same effect as rotating it about the x axis and then about the y axis. The obtained average temperature increase is $\Delta T = 212 \text{ K}$.

Looking at the Fig. S21c, the orientation response of the rod is studied. As in the case of the disk, a similar symmetry in the temperature increase distribution can be observed. The average temperature increase is remarkably lower than the one obtained in the disk, resulting in a minor tolerance to rotations, hence, showing a worse overall response.

In Fig. S21d, the case of the toroidal particle is analysed. The corresponding colormap exhibits weaker temperature increments for any ϕ value upon rotation in θ . Similarly to the disk, it offers a more stable thermal behaviour. Reduction of temperature increments is observed symmetrically for rotations in ϕ and θ , achieving a stable value of around 250 K for a full 90° rotation in either ϕ or θ . Attending unpolarized nature of the incident light beam, there always exists a configuration in which the electric field vector is parallel to the toroid transversal length, allowing the excitation of the corresponding mode. However, for each time in which the previous situation occurs, the opposite also takes place, with the electric field vector orthogonal to the toroid transversal length, avoiding the possible excitation mode. This leads to a commitment between both configurations, giving rise to a more stable excitation in time. The average temperature increment obtained is $\Delta T = 269$ K, proving that toroidal nanoparticles have a high tolerance to rotations.

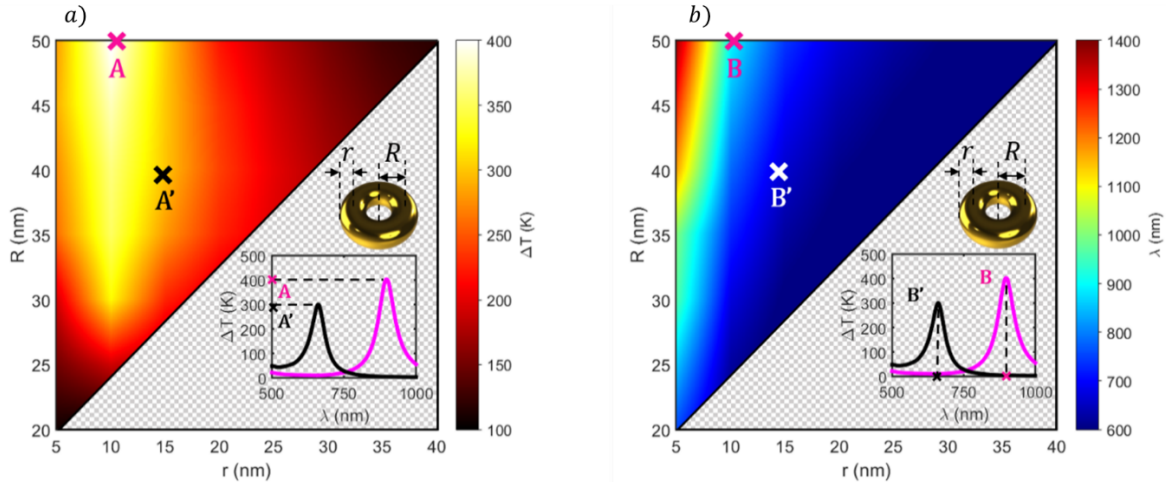


Fig. S22: Thermal response of a full syntonised toroidal nanoparticle in air. a) Maximum temperature increase of toroid for main (R) and secondary (r) radius variations represented between 0 and 400 K. b) Corresponding excitation wavelength of the maximum thermal response as a function of the main and secondary radii. The points A and B correspond to the spectral maximum temperature increase (inset) for the toroid shown in Fig. S20, $R=50$ nm and $r=10$ nm. Another example is given by A' and B' .

Fig. S22 displays two colormaps that correspond to the temperature increment produced by the toroid in air (Fig. S22a) and the wavelength at which the optimal resonance occurs for each case (Fig. S22b). Regarding design, main radii lower than 50 nm, more fitting to current nanoparticle sizes, have been considered. In addition, a secondary radius bigger than 5 nm has been assumed, as in the case of Fig. 4.

In sight of Fig. S22a, the tendency that both maximum temperature increases and excitation wavelength exhibit with respect to the structure dimensions can be observed. Note that the optimal temperature is reached when the secondary radius is fixed to 10 nm. On the one hand, larger secondary radii result in geometries closer to a sphere, and so does the temperature response. On the other hand, secondary radii smaller than 10 nm would lead to reduced toroid cross-sections, hindering the support of strong electromagnetic modes that can produce high thermal responses. Looking at the main radius, however, it can be observed that in general the temperature increments grow with it, as the toroid length is increased, which enhances the structure electrical resistance and its corresponding resistive losses.